SEQUENCE 4

Real numbers and intervals

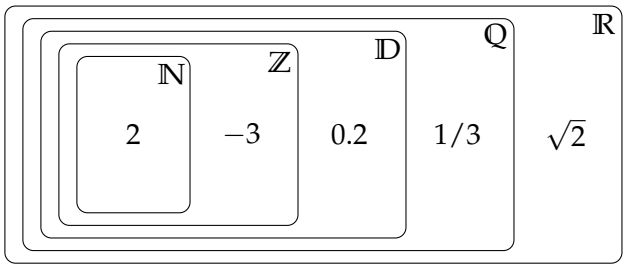
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4.1 Sets of numbers

Key point · Definitions.

- $\mathbb{N}$: natural numbers; $\mathbb{Z}$: integers.
- $\mathbb{D}$: decimal numbers, of the form $a/10^n$ with $a \in \mathbb{Z}$, $n \in \mathbb{N}$.
- $\mathbb{Q}$: rational numbers, of the form a/b with $a, b \in \mathbb{Z}$, $b \neq 0$.
- $\mathbb{R}$: real numbers, represented by the abscissae of the points of a graduated line. A real number that is not rational is **irrational**.

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{D} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}.$$



Example 1. Give the smallest of the five sets containing each of the numbers: -95; 7.29; $5/3$; $\sqrt{25}$; $\sqrt{2}$; π; 0.0002.

Solution. $-95 \in \mathbb{Z}$; $7.29 \in \mathbb{D}$; $5/3 \in \mathbb{Q}$; $\sqrt{25} = 5 \in \mathbb{N}$; $\sqrt{2}, \pi \in \mathbb{R} \setminus \mathbb{Q}$; $0.0002 = 2/10^4 \in \mathbb{D}$. The irrationality of π is accepted here.

4.2 Two proofs by contradiction

Method · Proof by contradiction. To prove a statement, assume its negation, deduce a contradiction, then conclude that the statement is true.

Key point · Property. $\frac{1}{3}$ is not a decimal number.

Proof to write.

Solution. Suppose $1/3 = a/10^n$ with $a \in \mathbb{Z}$, $n \in \mathbb{N}$. Then $10^n = 3a$ would be a multiple of 3. Yet $10 \equiv 1$ modulo 3, so 10^n leaves remainder 1 on division by 3 (the divisibility rule for 3 can also be used). Contradiction. Hence $1/3 \notin \mathbb{D}$.

Key point · Property. $\sqrt{2}$ is irrational.

Proof to write.

Solution. Suppose $\sqrt{2} = p/q$ with $p, q \in \mathbb{N}$, $q \neq 0$, and p/q in lowest terms. Then $p^2 = 2q^2$, so p^2 is even. If p were odd, its square would be odd: so p is even. Write $p = 2k$. We get $4k^2 = 2q^2$, then $q^2 = 2k^2$; so q is even in turn. Thus 2 divides p and q, contradicting irreducibility. Hence $\sqrt{2} \notin \mathbb{Q}$.

Teacher note. Have students identify separately the starting hypothesis, the implication used by contraposition on parity, and the final contradiction. Do not present a calculator's decimal expansion as a proof of irrationality.

4.3 Intervals of the real line

Key point · Notation. For two real numbers $a < b$, intervals describe the real numbers lying between two endpoints, with or without the endpoints. A square bracket includes the endpoint, a parenthesis excludes it. The symbols $-\infty, +\infty$ are not real numbers and always take a parenthesis.

Condition	Interval	Description
$a \leq x \leq b$	$[a, b]$	both endpoints included
$a < x \leq b$	$(a, b]$	a excluded, b included
$a \leq x < b$	$[a, b)$	a included, b excluded
$a < x < b$	(a, b)	both endpoints excluded
$x \geq a$	$[a, +\infty)$	greater than or equal to a
$x > a$	$(a, +\infty)$	strictly greater than a
$x \leq b$	$(-\infty, b]$	less than or equal to b
$x < b$	$(-\infty, b)$	strictly less than b

$$\mathbb{R} = (-\infty, +\infty), \quad \mathbb{R}^* = (-\infty, 0) \cup (0, +\infty).$$

Watch out. The strictly negative real numbers form $(-\infty, 0)$: 0 is excluded. $[a, a] = \{a\}$; $(a, a) = \varnothing$.

Example 2. Write as an interval then represent: $2 \leq x \leq 5$; $x > 4$; $-6 < x \leq 2$; $x < 0$.

Solution. $[2, 5]$; $(4, +\infty)$; $(-6, 2]$; $(-\infty, 0)$. On the number line, mark the endpoints, then show each one as included or excluded.

4.4 Intersection and union of intervals

Key point · Definition. $I \cap J$ corresponds to the condition "belonging to I **and** to J"; $I \cup J$ corresponds to "belonging to I **or** to J". A union of intervals is not always an interval.

Example 3. With $I = [0, 12]$ and $J = [3, 20]$, determine $I \cap J$ and $I \cup J$.

Solution. $I \cap J = [3, 12]$ and $I \cup J = [0, 20]$.

Example 4. Complete, justifying with representations on a number line.

I	J	$I \cap J$	$I \cup J$
$(-4, 3]$	$[1, 5]$	$[1, 3]$	$(-4, 5]$
$(-\infty, 2]$	$[-4, +\infty)$	$[-4, 2]$	$\mathbb{R}$
$(-\infty, 7]$	$[7, +\infty)$	$\{7\}$	$\mathbb{R}$
$(-\infty, 5]$	$[-3, +\infty)$	$[-3, 5]$	$\mathbb{R}$
$(-2, 0)$	$(0, 3]$	$\varnothing$	$(-2, 0) \cup (0, 3]$

Solution. For each intersection, keep the points common to both representations; for each union, keep all the points of at least one. In the last row, 0 belongs to neither interval: the union cannot be replaced by $(-2, 3]$.

Fluency · 5–10 min. Simplify $\overrightarrow{AB} + \overrightarrow{BC}$; solve $3x - 6 = 0$; calculate 10^{-2}.

Answer. $\overrightarrow{AC}$; $x = 2$; 0.01.

4.5 Absolute value and distance

Key point · Definition and properties. $|x|$ is the distance from the real number x to 0; $|x| \geq 0$ and

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

The distance between a and b is $|b-a|$. For $a \in \mathbb{R}$ and $r \geq 0$:

$$|x-a| \leq r \iff x \in [a-r, a+r], \qquad |x-a| < r \iff x \in (a-r, a+r).$$

Example 5. Calculate $|3.2|$, $|-1.6|$, $|\sqrt{3}-2|$, then the distances between 2 and 5, 2 and -5, -2 and -5.

Solution. 3.2; 1.6; $2-\sqrt{3}$ since $\sqrt{3} < 2$. The distances are 3, 7 and 3.

Example 6. Solve: $|x-7/3| = 14$; $|x-2| \leq 3$; $|x+3| < 5$.

Solution. For the first equation, $x - 7/3 = 14$ or $x - 7/3 = -14$, so $S = \{-35/3, 49/3\}$. Then $|x-2| \leq 3 \iff x \in [-1, 5]$. Finally $|x+3| = |x-(-3)| < 5 \iff x \in (-8, 2)$.

Example 7. Write $x \in [4, 10]$ as an inequality with an absolute value.

Solution. The centre is $(4+10)/2 = 7$ and the radius $(10-4)/2 = 3$: $|x-7| \leq 3$.

4.6 Bounds and approximations

Key point · Vocabulary. To bound x is to give $a \leq x \leq b$; the width is $b - a$. An approximation from below is less than or equal to x; an approximation from above is greater than or equal to it. Rounding chooses the nearest value at the required precision. Significant figures are counted from the first nonzero digit; the displayed precision must suit the context.

Example 8. Bound $\sqrt{2}$ between two consecutive hundredths. Give its rounding to the nearest hundredth. Round 0.004567 to three significant figures, then 12345 to three significant figures in scientific notation.

Solution. $1.41^2 = 1.9881 < 2 < 2.0164 = 1.42^2$, so $1.41 < \sqrt{2} < 1.42$ (width 0.01). The rounding to the nearest hundredth is 1.41. The other roundings are 0.00457 and 1.23×10^4.

```
def encadre_racine2(p):
    # p entier naturel ; calculs entiers exacts
    q = 10**p
    k = 0
    while (k+1)**2 <= 2*q*q:
        k = k+1
    return k/q, (k+1)/q
```

Example 9. Explain why this function bounds $\sqrt{2}$. What does it return for $p = 2$? Why does it terminate?

Solution. The loop looks for the largest integer k such that $k^2 \leq 2q^2$. On exit, $k^2 \leq 2q^2 < (k+1)^2$, so $k/q \leq \sqrt{2} < (k+1)/q$. For $p = 2$, it returns $(1.41, 1.42)$. Since $(2q)^2 > 2q^2$, the counter eventually reaches a value that makes the test false. The final divisions are floating-point approximations; the square tests are exact.